Tutorial 11-6

Use ModelBuilder for a risk index

People who live in poverty often have poor health care, unhealthy diets, and unhealthy habits such as smoking—all factors contributing to heart attacks. In this tutorial, you will create an index for identifying poverty areas by combining four poverty indicators[2]: population (1) below the poverty income line, (2) of female-headed households with children, (3) with less than a high school education, and (4) of workforce males who are unemployed.

Robyn Dawes provides a simple method for combining such measures into a poverty index[3]. If you have a reasonably good theory that several variables are indicative or predictive of a dependent variable of interest (and whether the dependent variable is observable or not), then Dawes makes a good case that all you need to do is to remove scale from each input, so each has the same weight, and then average the scaled inputs to create a predictive index. A good way to remove scale from a variable is to calculate z-scores, subtracting the mean and then dividing by the standard deviation for each variable.

You can see in the following table that if you simply averaged the four variables, then Male unemployed arbitrarily gets the highest weight while female-headed households would have practically no weight, given the means of the variables. Z-scores for all four variables, however, all have means of zero and standard deviations of one, so when averaged they each will have equal weight.

Indicator variable	Mean	Standard Deviation
Female-headed households with children	1.422	4.431
Less than high school education	110.060	80.812
Male unemployed	154.500	124.804
Poverty income	126.021	147.188

There are three parts to creating the poverty index. First you will calculate the z-scores for each of the four indicators. The map layers for the indicators are centroids of blocks for the population of female-headed households with children and centroids of block groups

2 W. O'Hare and M. Mather, "The growing number of kids in severely distressed neighborhoods: Evidence from the 2000 census," *Kids Count* (2003). *Kids Count* is a publication of the Annie E. Casey Foundation and the Population Reference Bureau, and is available from `http://www.aecf.org/upload/publicationfiles/da3622h1280.pdf`.

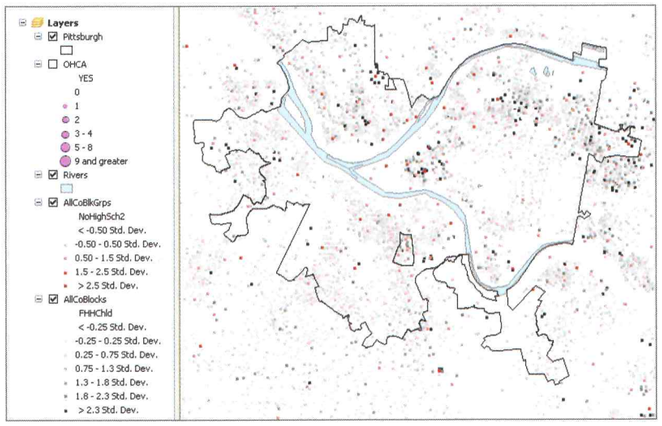
3 R. M. Dawes, "The robust beauty of improper linear models in decision making," *American Psychologist* 34 (1979): 571–582.

for the other three indicators (which are not available at the more desirable, smaller block level). Thus to make these layers comparable for combining them into an index, you will transform them into kernel density maps, all with the same 150-foot-square grid cells. So the second part is to create kernel density maps for all four input variables. The third part is to use a Spatial Analyst tool to add the surfaces, weighted by 0.25, to average them. You will carry out parts 2 and 3 using ModelBuilder to document the work and provide a reusable tool for creating an index. Note that you can work through the following exercises successfully even if you did not complete the introduction to ModelBuilder in tutorial 8-7. Tutorial 8-7 has a more complete introduction to ModelBuilder.

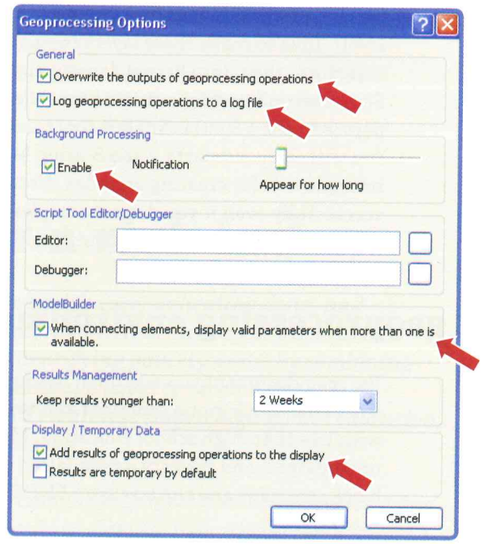
Set geoprocessing environment

The map document you will open has inputs for preparing the poverty index: AllCoBlkGrps, which has block group centroids and needed attributes (NoHighSch2 = population with less than high school education, Male16Unem = males in the workforce who are unemployed, and Poverty = population below poverty income), and AllCoBlocks, which has block centroids and the attribute FHHChld = female-headed households with children.

1 **In ArcMap open Tutorial11-6.mxd from the \ESRIPress\GIST1\Maps\ folder.** Shown are the block group centroids and block centroids, each displaying one of the four poverty indicators via a color ramp. You can see that it is difficult to represent the spatial patterns effectively using vector graphics, plus it is difficult to integrate the information from just two spatial distributions out of the four needed for the poverty index. The raster poverty index that you will create will do a better job on both issues.



2 On the main menu, click Geoprocessing, Geoprocessing Options, and make selections as shown in the image at the right.



3 Click OK.

4 On the main menu, click Geoprocessing, Environments, Raster Analysis, and Select As Specified Below for Cell Size. Type **150** for the specification, select Pittsburgh for the Mask, and click OK.

5 Save the map document in \ESRIPress\GIST1\MyExercises\Chapter11\.

Standardize input variables

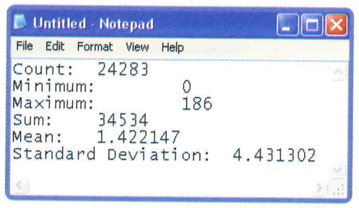
Here you will calculate the z-score in an attribute table of one of the input feature classes. To save time, the other three variables already have z-scores ready for use.

1 Right-click AllCoBlocks in the TOC and click Open Attribute Table.

2 Scroll to the right, right-click FHHChld, and click Statistics. It is convenient to copy and paste the statistics to Notepad and then copy and paste them later to the field calculator that you will use.

3 On your desktop, click Start, All Programs, Accessories, Notepad.

4 Select all of the statistics in the Statistics of AllCoBlocks window, pres Ctrl+C, click inside the Notepad window, and press Ctrl+V.



5 Close the Statistics of AllCoBlocks window, click the Table Options button in the Table window, click Add Field, type **ZFHHChld** for Name, select Float for Type, and click OK.

6 Right-click the column heading for ZFHHChld, click Field Calculator, and create the following expression in the bottom panel of the Field Calculator window by copying and pasting from your Notepad window,:

([FHHChld] – 1.422147) / 4.431302

7 Click OK. To the right are the first six values for the calculated z-scores.

FHHChld	ZFHHChld
0	-0.320932
0	-0.320932
0	-0.320932
2	0.130403
4	0.581737
1	-0.095265

8 Close the table and Notepad.

Create a new toolbox and model

1 Click Windows and Catalog, and expand Home – Chapter 11 in the folder/file tree.

2 Right-click Home – Chapter 11, click New and Toolbox, and rename the new toolbox **UnweightedIndices.tbx**.

3 Right-click UnweightedIndices.tbx and click New, Model.

4 In the Model window, click Model, Model Properties.

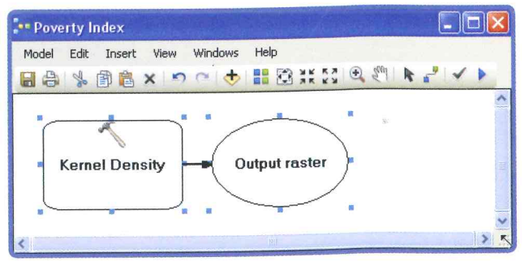
5 On the General tab, for Name type **PovertyIndex** (no spaces allowed), for Label type **Poverty Index**, click OK, and hide the Catalog window.

Create a kernel density layer for an input

The next task is to create kernel density layers for the four inputs using the z-scores. After you create model elements for one kernel density layer, you can easily copy it and make adjustments for the remaining three.

1 Click Windows, Search.

2 In the Search window, type **Kernel Density** in the search text box, press Enter, and drag Kernel Density to the Poverty Index model window and drop it in.



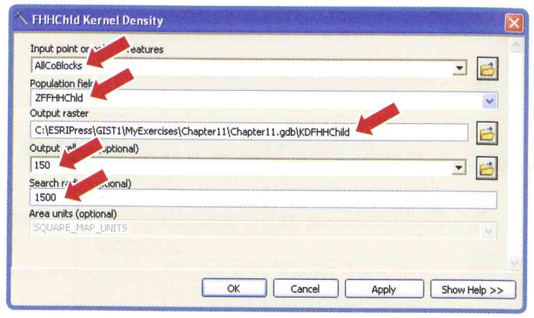
3 In the model, right-click Kernel Density and click Open.

4 Type or make the selections as shown in the image at the right.

5 Click OK.

6 Right-click the Kernel Density tool element, click Rename, and change the name to **FHHChld Kernel Density**.

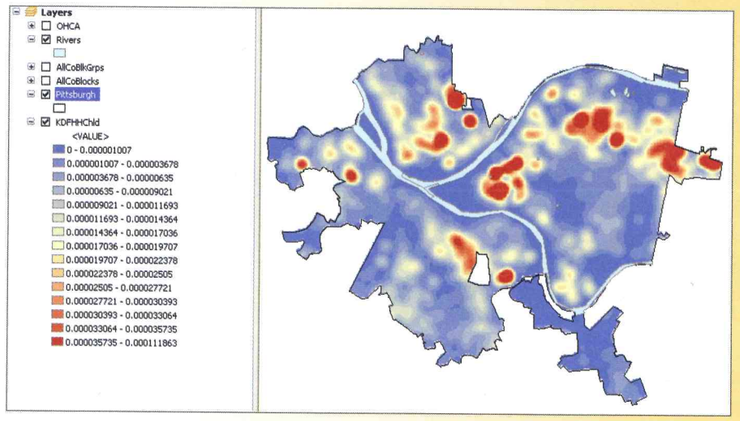
7 Right-click FHHChld Kernel Density and click Run.



8 Right-click the KDFHHChld and click Add to Display.

YOUR TURN

Resymbolize the new layer using the Classified method with 1/4 standard deviations and the color ramp that runs from blue to yellow to red. Turn off the point feature layers. The result is as follows:



Create a kernel density layer for a second input

You can reuse the model elements you just built. While blocks work very well with a search radius of 1,500 feet, there are fewer block groups (the remaining three poverty inputs are at the block group level), so you need a larger search radius of 3,000 feet for them.

1 In the Model window, right-click FHHChld Kernel Density and click Copy.

2 Click Edit, Paste.

3 Right-click the new FHHChld Kernel Density 2 model element and rename it **NoHighSchKernelDensity**.

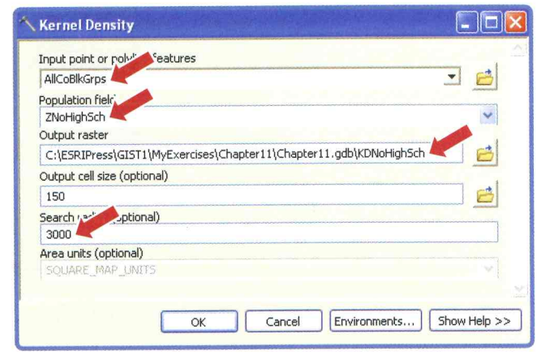
4 Right-click NoHighSch Kernel Density and click Open. Ignore the error messages. You will make changes that eliminate them.

5 Type or make the selections as shown in the image.

6 Click OK.

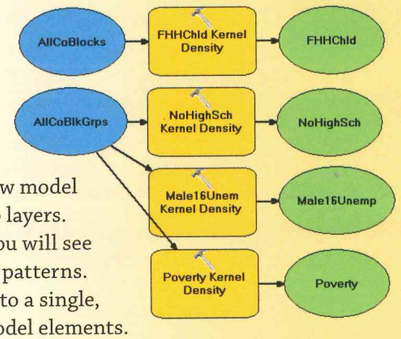
7 Right-click KDFHHChld(2) and rename it **NoHighSch**.

8 Right-click NoHighSch Kernel Density, click Run, and resymbolize as you like.



YOUR TURN

Copy and paste the NoHighSch Kernel Density model element two times to use block group attributes ZMaleUnem and ZPoverty to create two new raster layers. See the resulting partial model at right for element names that you need to use. Then run each of the two new model elements and resymbolize resulting map layers. Examine each of the four raster maps. You will see that they have overlapping but different patterns. The index will combine these patterns into a single, overall pattern. Resize and rearrange model elements. Run the three processes you just created. Resymbolize each new layer using standard deviations and color ramps of your choice. Notice that the process model elements acquire drop shadows in the model window after you run them. To reset the model so that you can run it again, if needed, click Model, Validate Entire Model. ModelBuilder removes the drop shadows. Save your model.

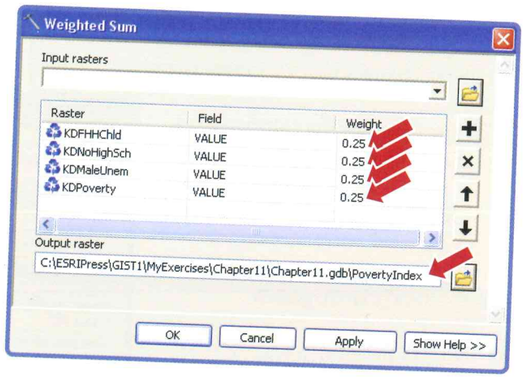


Average kernel density maps

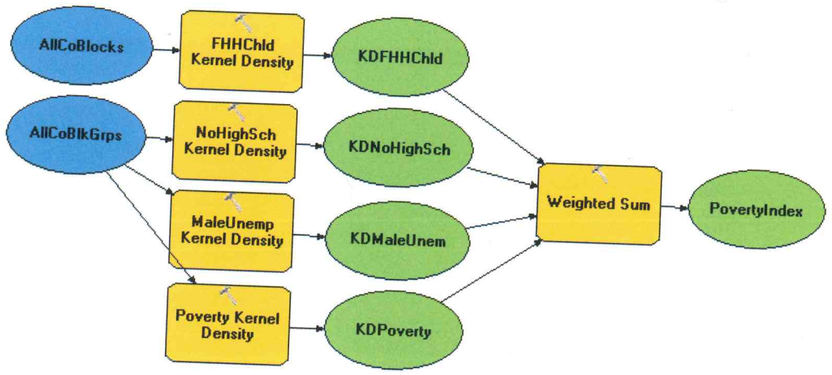
1 Type **Weighted Sum** in the search text box and press Enter.

2 Drag the Weighted Sum link to your model, to the right of the kernel density outputs, and drop it in.

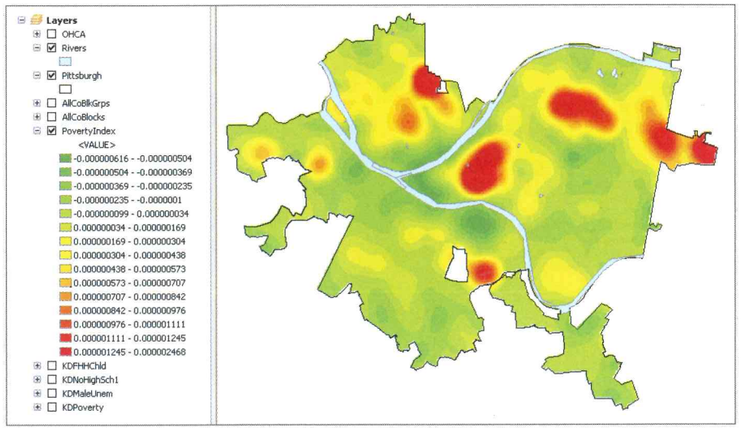
3 Right-click Weighted Sum, click Open, and type or make selections as shown in the image to the right.



4 Click OK and use the following model to complete renaming model elements.



5 Run the Weighted Sum process and resymbolize the resulting PovertyIndex using standard deviations and the green-to-yellow-to-red color ramp.



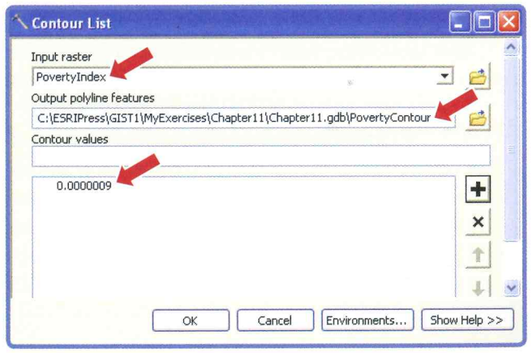
6 Save your model.

Create poverty contour

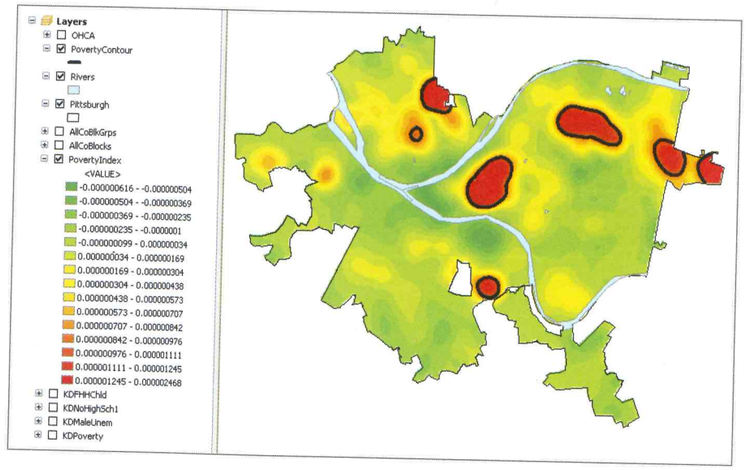
Suppose that after consideration, policy analysts wish to use the poverty index density of 0.0000009 or higher to define poverty. Next, you will create a feature class that has the contour line for that index "elevation."

1 In the search text box, type **Contour List** and click the Contour List link.

2 Type or make the selections as shown.



3 **Click OK.** You now have a set of polygons, shown with thick black outlines, that explicitly define poverty areas and can be used for many policy purposes.



4 **Save the map document and close ArcMap.**